

Akeli Adilijiang

Computer Science Student

Laguna Hills, CA

akil08141202@gmail.com

GitHub: [Akil0814](#)

Portfolio: [akil0814.github.io](#)

SUMMARY

Computer Science student focused on C++ graphics and software engineering. Experienced in building modular C++ projects with SDL, Qt, OpenGL, SQLite, and CMake, with an emphasis on software design and maintainability.

TECHNICAL SKILLS

Programming Languages: C++, C, Python, Lua, SQL, GLSL

Graphics & Rendering: OpenGL (GLFW, GLAD), SDL2/SDL3, GDI

Web: HTML, CSS, JavaScript

Databases & Data: SQLite, PostgreSQL, cJSON

Tools & Environments: Git, GitHub, CMake, vcpkg, Docker, Visual Studio, VS Code, MinGW, VMware, SecureCRT, Codex

Platforms: Windows, Linux (Ubuntu, CentOS)

Languages: English (Fluent), Chinese (Native), Japanese (Basic)

PROJECTS

SoftRenderer (C++ / Win32 / GDI) | [Overview](#)

- Built a basic 3D rendering interface from scratch without external graphics libraries.
- Implemented OpenGL-style features including texture sampling, perspective correction, depth testing, color blending, backface culling, and frustum clipping.
- Developed an ImGui-compatible rendering backend for performance monitoring and future optimization.

WarShip (C++ / SDL) | [Overview](#)

- Designed a modular architecture separating input, update, rendering, and state management to improve maintainability.
- Implemented asset/resource pooling and object pooling for scene entities to reduce repeated loading and runtime allocations.
- Built a callback-driven VFX/animation system and entity factories to keep gameplay logic decoupled.

PathFindingVisualizer (C++ / SDL / Algorithm) | [Overview](#)

- Built a pathfinding visualizer with dynamic wall editing, algorithm switching, and step playback.
- Published a web demo; used for a data structures course demonstration and received 20+ GitHub stars.

Take Me to Ballgame (C++ / Qt / SQLite) | [Overview](#)

- Implemented a modular backend core with repository, service, model, and application layers.
- Built SQLite schema checks, transactional CSV import, and backup/rebuild/restore flows.
- Implemented module-level fallbacks to keep core features usable when optional data was unavailable.

8-bit CPU / Digital Logic Build (74xx TTL) | [Overview](#)

- Built and wired CPU-style modules on breadboards using 74-series TTL logic.
- Designed a 16x8 memory architecture with an address register and program counter for instruction sequencing.

EDUCATION

Saddleback College | Computer Science (in progress) | GPA: 3.9

Relevant Coursework: Data Structures, Physics (E&M), Linear Algebra, Calculus

Expected Transfer: 9/2027

JOB EXPERIENCE

UPS Preload

September 2023 – August 2025

- Sorted 1,200+ packages per shift under strict early-morning deadlines while maintaining accuracy.
- Balanced full-time coursework with shifts regularly starting before 4:00 AM, demonstrating reliability and time management.

Saddleback College – CARE Corner Assistant

October 2025 – Present

- Assisted an average of 70+ students daily with campus resource navigation, service referrals, and administrative support.
- Coordinated basic-needs supplies, check-ins, and front-desk support for daily student-service operations.

ACTIVITIES

CES – Exhibitor | Las Vegas, NV | 2026

- Presented product demos and communicated features and technical highlights to attendees.